Notes on Bloom et al. (2018)

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Motivation Uncertainty is importqnt. Try to evaluate the role of uncertainty for business cycle.

Measurement A firm produces output according to

$$y_{j,t} = A_t z_{j,t} f(k_{j,t}, n_{j,t})$$

where A_t is the agggregate productivity and $z_{j,t}$ is the idiosyncratic component. They follow an $AR(1)$ process:

$$\log A_t = \rho^A \log A_{t-1} + \sigma^A_{t-1}\epsilon_t$$
$$\log z_{j,t} = \rho^z \log z_{j,t-1} + \sigma^z_{t-1}\epsilon_{j,t}.$$

Let the variances move over time according to two-state Markov chains.

Empirics Uncertainty is countercyclical, but the causality seems to be two-way.

(a) Dispersion in plant-level innovation is countercyclical.

  (a) Data: Census panel of manufacturing establishments.

  (b) Calculate the establishment-lebel TFP $\hat{z}_{j,t}$ according to Foster, Haltiwanger and Krizan (2000). Define TFP **shock** as $e_{j,t}$:

$$\log \hat{z}_{j,t} = \rho \log \hat{z}_{j,t-1} + \underbrace{\mu_j}_{\text{establishment FE}} + \underbrace{\lambda_t}_{\text{time FE}} + e_{j,t}.$$

Define microeconomic uncertainty $\sigma^{\hat{z}}_{t-1}$ as the cross-sectional dispersion of $e_{j,t}$ calculated on an annual basis.

(c) Empirical strategy: Regress dispersion measurements (standard deviation, skewness, kurtosis, and IQR[1]) on the number of recession quarters in a year (*e.g.*, recession only in Q4 will be assigned with number 0.25).

(b) Industry-level uncertainty is countercyclical.

(c) Macroeconomic measures of uncertainty: Estimated using GARCH(1,1) model on Basu, Fernald and Kimball (2006) (seems to have been mentioned in Byoungchan's class).

Model

References

Bloom, Nicholas, Max Floetotto, Nir Jaimovich, Itay Saporta-Eksten, and Stephen J. Terry, "Really Uncertain Business Cycles," *Econometrica*, 2018, *86* (3), 1031–1065.

[1]Interquartile range, which is the upper quartiel minus the lower quartile.